

1. PROJECT OVERVIEW

- **Project Title:** FitBuddy - AI Workout Generator

2. BACKEND & SERVER ARCHITECTURE

- **Core Language:** Python 3
- **Web Framework:** FastAPI - Chosen for its modern, high-performance capabilities in handling asynchronous web requests and routing.
- **ASGI Server:** Uvicorn - The lightning-fast server utilized to run the FastAPI application, serving as the bridge between the backend logic and the web.

3. DATABASE & ORM LAYER

- **Database Engine:** SQLite - A lightweight, file-based relational database ideal for rapid prototyping and local storage of user metrics.
- **Object-Relational Mapping (ORM):** SQLAlchemy - Used to translate Python classes (like User and WorkoutPlan) into SQLite database tables, abstracting complex SQL queries.

4. ARTIFICIAL INTELLIGENCE LAYER

- **LLM Provider:** Google Gemini API.
- **Specific Model:** gemini-3.5-flash-lite - Specifically selected for its rapid text-generation speed, ensuring the user does not wait long for their 7-day schedule to be generated.

5. FRONTEND PRESENTATION TIER

- **Markup & Styling:** HTML and CSS - Used to construct a clean, responsive web interface without the heavy bloat of JavaScript frameworks.
- **Templating Engine:** Jinja2 Templates - Interfaced directly with FastAPI to dynamically inject the backend AI strings (the workout plan) directly into the frontend HTML blocks.